

Chapter 27 Fermi and Bose Statistics

Chapter 13 ~ classical statistics or kinetic theory,
the fundamental laws of motion ~ Newton's laws.

system of 'ideal' classical particles

probability of finding a particle in $\begin{matrix} v_x \rightarrow v_x + dv_x \\ v_y \rightarrow v_y + dv_y \\ v_z \rightarrow v_z + dv_z \end{matrix}$ is

$$f(v_x, v_y, v_z) dv_x dv_y dv_z \\ = \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left\{ - \frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right\} dv_x dv_y dv_z$$

Maxwell distribution

Maxwell distribution is a special case of Boltzmann distribution:

$$f(p, q) dp dq = \frac{e^{-\beta E(p, q)} dp dq}{\iint e^{-\beta E(p, q)} dp dq} \quad \left(\beta = \frac{1}{k_B T} \right)$$

Microscopic particles are essentially **quantal**

Classical or quantal?

define a thermal (de Broglie) wavelength

$$\lambda = \frac{h}{mv_{\text{rms}}} = \frac{h}{m} \sqrt{\frac{m}{3k_B T}}$$

$$\lambda = \frac{h}{mv_{\text{rms}}} = \frac{h}{m} \sqrt{\frac{m}{3k_{\text{B}}T}}$$

a — average separation of particles.

$$a > \lambda, \quad \text{or} \quad T > \frac{m}{3k_{\text{B}}} \left(\frac{h}{ma} \right)^2 \equiv T_0$$

granular property dominant

Degenerate temperature

(The classical Boltzmann statistics can be used.)

$$a < \lambda \quad \text{or} \quad T < T_0$$

wave property is more important — quantal

Identical particles are essentially indistinguishable.

$$\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots, \mathbf{r}_N) = \mp \psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N)$$

anti-symmetric — fermions

symmetric — bosons

Particles with odd half integer spin are fermions.

electrons, protons, neutrons, muons...

Particles with integer spin are bosons.

Photons, pions, mesons...

The aggregates of arbitrary number of bosons and even number of fermions are bosons.

The aggregates that consist odd number of fermions are fermions.

Bosons: ^1H , nuclei of ^2H (deuteron), nuclei of ^4He , ^4He atoms etc.

Fermions: ^2H (deuterium), nuclei of ^3H (tritium), and ^3He atoms etc.

^1H (protium)

The density of states

For size of macroscopic scale energy levels are quasi-continuous.

1D:



$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E \psi(x)$$

$$\frac{d^2}{dx^2} \psi(x) + k^2 \psi(x) = 0, k^2 = \frac{2mE}{\hbar^2}.$$

$$\psi\left(-\frac{L}{2}\right) = \psi\left(\frac{L}{2}\right) \quad \text{periodic boundary condition}$$

$$\psi(x) = Ae^{ikx} \longrightarrow Ae^{-ik\frac{L}{2}} = Ae^{ik\frac{L}{2}},$$

$$e^{ikL} = 1.$$

$$k_n = \frac{2n\pi}{L}, n = 0, \pm 1, \pm 2, \dots$$

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2}{2m} \left(\frac{2n\pi}{L} \right)^2.$$

In k -space (1D), interval $\frac{2\pi}{L} \leftrightarrow 1$ state

3D case:

$$E_{n_x, n_y, n_z} = \frac{\hbar^2}{2m} \left(\frac{2\pi}{L} \right)^2 (n_x^2 + n_y^2 + n_z^2)$$

$$= \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2)$$

$$= \frac{\hbar^2 k^2}{2m}.$$

$$k_i = \frac{2n_i\pi}{L}, \quad i = x, y, z, \quad n_i = 0, \pm 1, \pm 2, \dots$$

In k -space (3D), a cube of volume $\left(\frac{2\pi}{L} \right)^3 \leftrightarrow 1$ state.

For one-dimensional case

$$E_n = \frac{\hbar^2}{2m} \left(\frac{2\pi}{L} \right)^2 n^2 \equiv E_1 n^2$$

$$E_1 = \frac{1}{2mc^2} \left(\frac{hc}{L} \right)^2$$

$$\begin{aligned} &\sim \frac{1}{2 \times 0.511 \text{ MeV}} \left(\frac{12.400 \text{ keV} \cdot \text{\AA}}{10^{-2} \text{ m}} \right)^2 & L \sim 10^{-2} \text{ m} \\ &= 1.50 \times 10^{-14} \text{ eV} \end{aligned}$$

quasi-continuity: summation over energy levels

→ integral

$$\sum_i (\dots) \longrightarrow gV \int (\dots) \frac{d^3k}{(2\pi)^3}$$

$g = 2s + 1$
degeneracy due to spin

For non-relativistic system, energy spectrum is

$$\mathcal{E} = \frac{\hbar^2 k^2}{2m}.$$

Then

$$gV \frac{4\pi k^2}{(2\pi)^3} dk = gV \frac{4\pi k}{(2\pi)^3} d\left(\frac{k^2}{2}\right) = gV \frac{4\pi mk}{(2\pi)^3 \hbar^2} d\left(\frac{\hbar^2 k^2}{2m}\right)$$

$$= \frac{gV}{4\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon^{1/2} d\varepsilon = \underline{g(\varepsilon)} d\varepsilon$$

$$k = \frac{\sqrt{2m\varepsilon}}{\hbar}$$

density of states (DOS)

$$\sum_i (\cdots) \longrightarrow \int (\cdots) g(\varepsilon) d\varepsilon$$

27.2 Fermi-Dirac distribution

For ideal fermi gases, the occupation number on the energy level E_j

$$n_j^F = \frac{g_j}{e^{\beta(E_j - \mu)} + 1}$$

the degeneracy of the energy level

$$\beta \equiv 1/k_B T$$

the chemical potential

$$f_{FD}(E_j) = \frac{n_j^F}{g_j} = \frac{1}{e^{\beta(E_j - \mu)} + 1}.$$

the Fermi-Dirac function

The chemical potential is determined by

$$\sum_j n_j^F = \sum_j \frac{g_j}{e^{\beta(E_j - \mu)} + 1} = N,$$

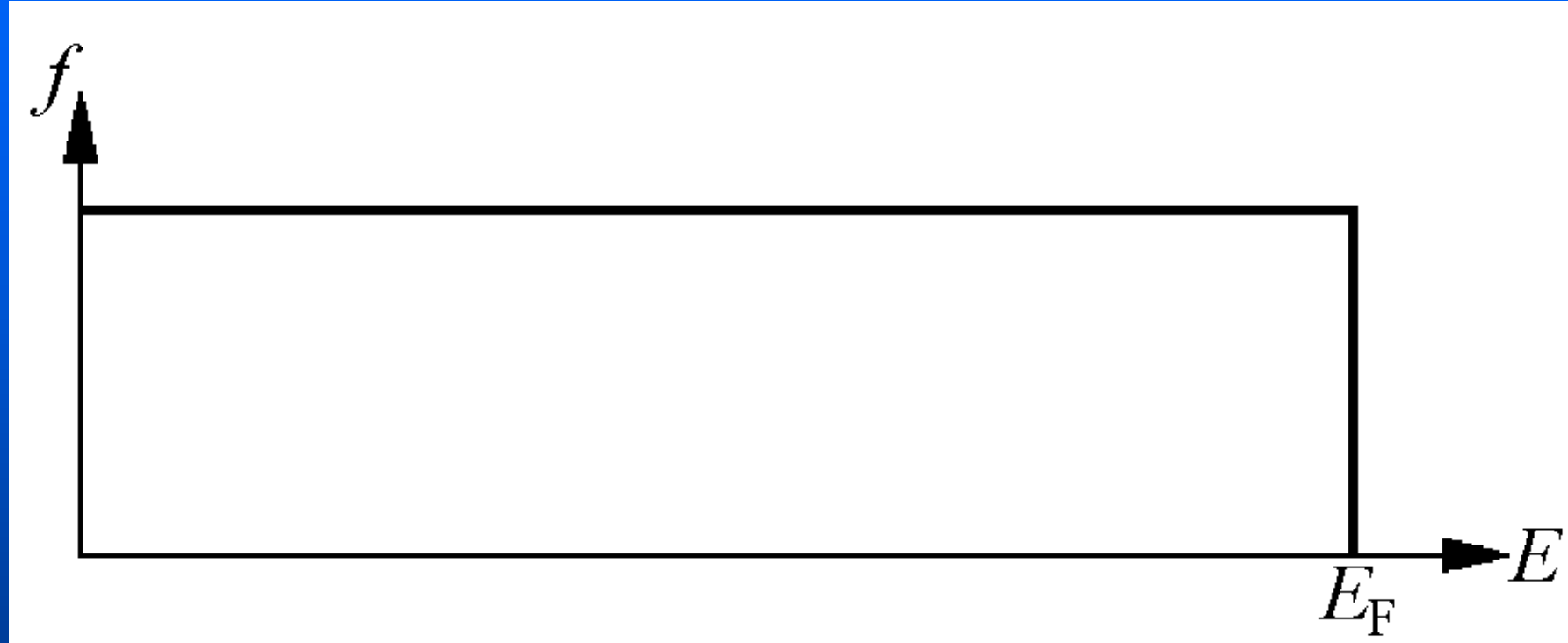
where N is the total number of particles.

For $T=0, \mu = E_F$ (the Fermi energy, the Fermi level)

$$f_{FD}(E_i) = \frac{1}{e^{\beta(E_j - \mu)} + 1}$$



$$f_{FD}(E) = \theta(E_F - E) = \begin{cases} 1 & E < E_F \\ 0 & E > E_F \end{cases}$$



Fermi wavevector and Fermi temperature

$$E_F = \frac{\hbar^2 k_F^2}{2m} \equiv k_B T_F$$

In k -space, particles only fill a zone $k \leq k_F$ called Fermi sphere.

$$\begin{aligned} N &= (2s+1)V \int \frac{d^3k}{(2\pi)^3} = \frac{2V}{(2\pi)^3} \int_0^{k_F} 4\pi k^2 dk \\ &= \frac{V}{3\pi^2} k_F^3 \end{aligned}$$

Fermi wave number

$$k_F = \left(3\pi^2 \frac{N}{V} \right)^{1/3}$$

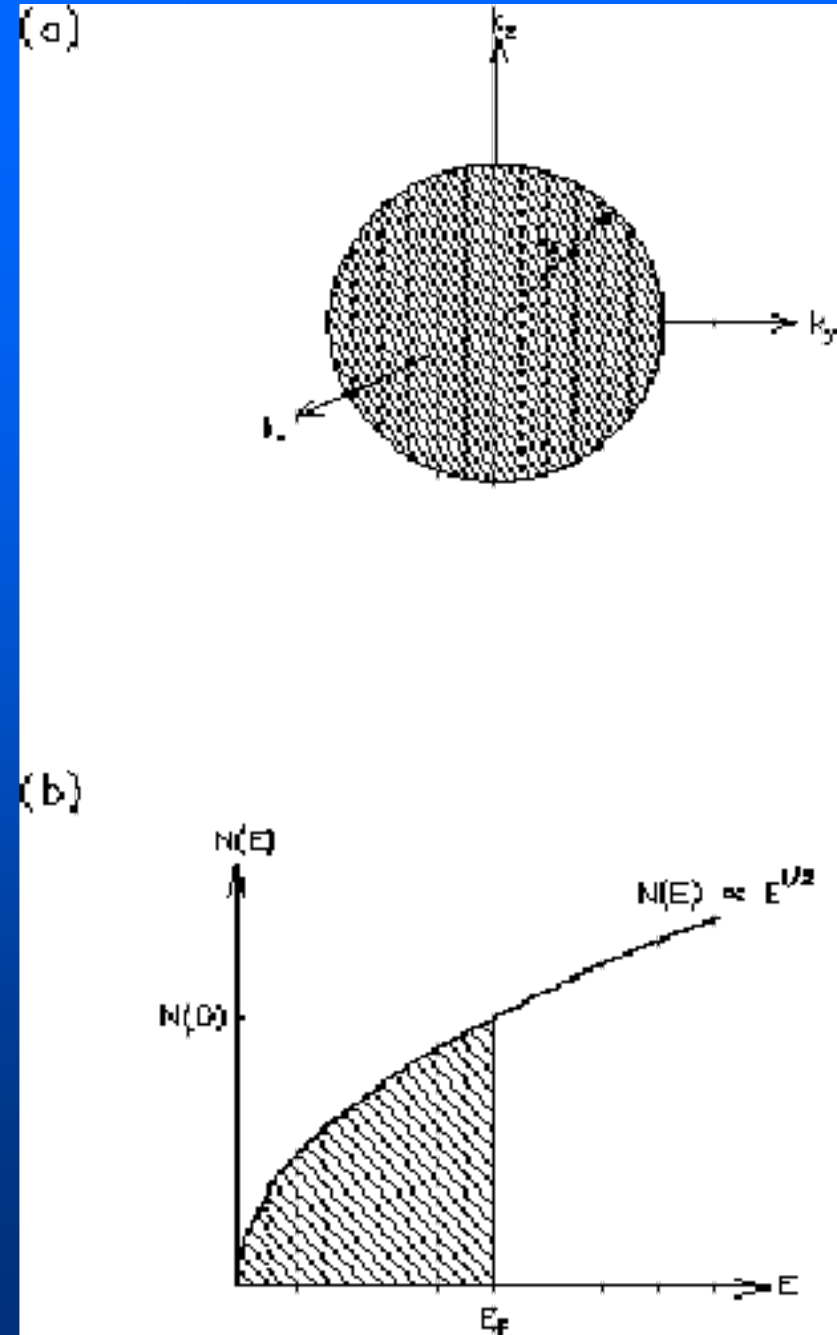
For metal copper

$$n = 8.45 \times 10^{28} / \text{m}^3 ,$$

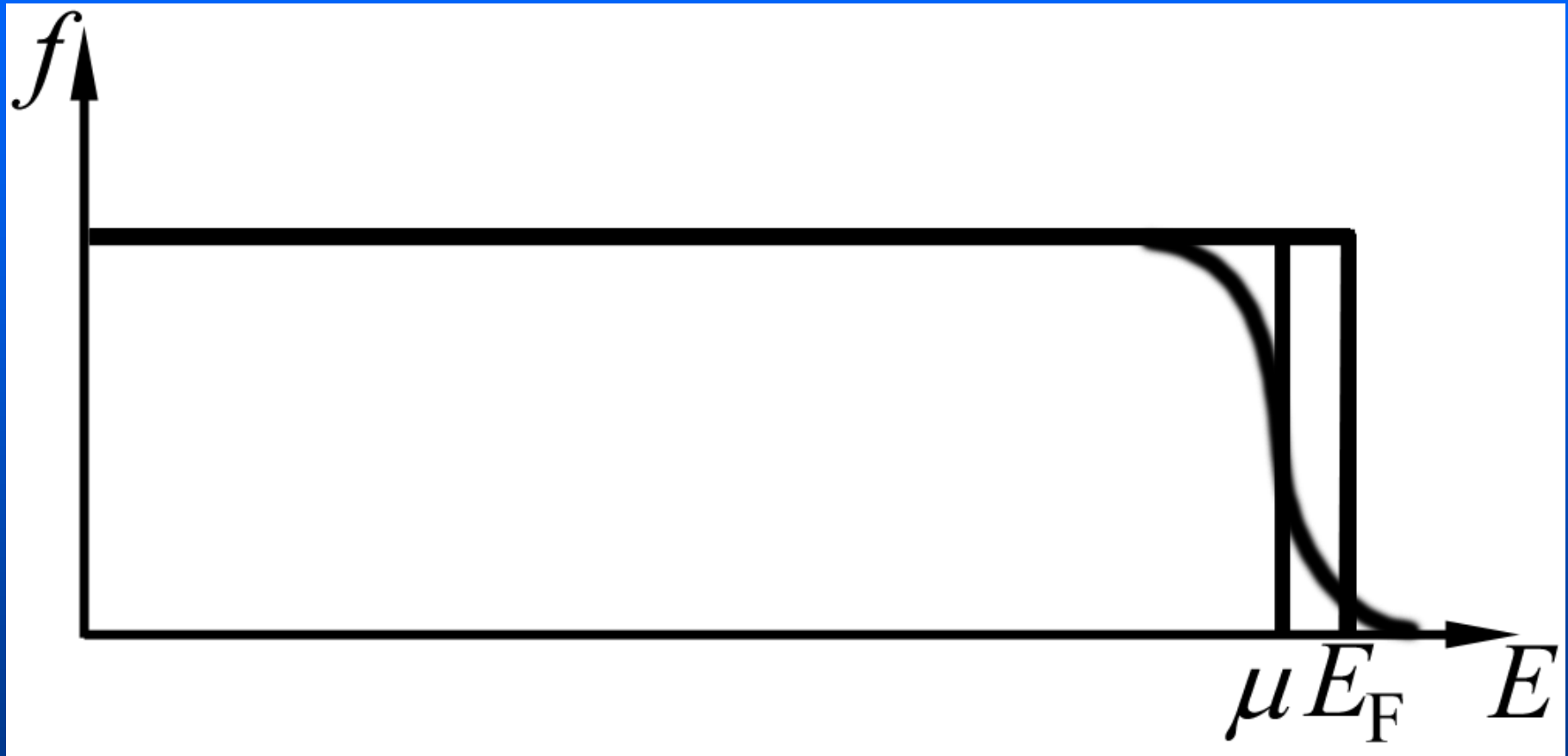
$$k_F = 1.36 \times 10^{10} / \text{m} ,$$

$$E_F = 1.12 \times 10^{-18} \text{ J} = 7.00 \text{ eV}$$

$$T_F = 8.12 \times 10^4 \text{ K}$$



$$k_{\text{B}}T \sim 300\text{K} \sim 0.025\text{eV}$$



$$\begin{aligned}
\frac{\langle E \rangle}{N} &= \frac{\int_0^{\infty} E \theta(E_F - E) g(E) dE}{\int_0^{\infty} \theta(E_F - E) g(E) dE} \\
&= \frac{\int_0^{\infty} E \theta(E_F - E) \sqrt{E} dE}{\int_{E_{F0}}^{\infty} \theta(E_F - E) \sqrt{E} dE} \\
&= \frac{\int_0^{E_F} E \sqrt{E} dE}{\int_0^{E_F} \sqrt{E} dE} = \frac{3}{5} E_F
\end{aligned}$$

Assignment: 27.2, 27.4

27.3 Bose-Einstein distribution

Bose-Einstein distribution

$$n_j^B = \frac{g_j}{e^{\beta(E_j - \mu)} - 1}$$

$$f_{BE}(E_j) = \frac{n_j^B}{g_j} = \frac{1}{e^{\beta(E_j - \mu)} - 1} \quad \text{Bose-Einstein function}$$

$$f_{BE}(E_j) \geq 0 \Rightarrow \mu \leq 0. \quad (\text{Assume the lowest energy level } E_0=0.)$$

For the system that the total number of particles is not constant, such as photon gas, we have $\mu=0$.

Blackbody radiation

The photon gas $\mu=0$

$$n_j^B = \frac{g_j}{e^{E_j/k_B T} - 1}$$

$$dn = \frac{g(E)dE}{e^{E/k_B T} - 1}$$

The number of standing waves(states) is inside the shell of thickness dk in the k -space is



$$2 \times \frac{4\pi k^2 dk}{\left(\frac{2\pi}{L}\right)^3} = 2 \times V \frac{k^2 dk}{2\pi^2}$$
$$= \frac{8\pi V}{c^3} \nu^2 d\nu$$
$$k = \frac{2\pi\nu}{c}$$

An arrow points from the $k^2 dk$ term in the first equation to the $k = \frac{2\pi\nu}{c}$ equation.

$$E = h\nu \rightarrow g(E)dE = g(\nu)d\nu = \frac{8\pi V}{c^3} \nu^2 d\nu$$

$$dn = \frac{8\pi\nu^2 V}{\underline{c^3}} \frac{1}{e^{h\nu/k_B T} - 1} \underline{d\nu}$$

$$u(\nu, T) d\nu = \frac{h\nu dn}{V} = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{h\nu/k_B T} - 1} d\nu$$

This is the celebrated Planck radiation formula.

$$u(T) = \frac{U(T)}{V} = \int_0^\infty u(\nu, T) d\nu$$

$$= \int_0^\infty \frac{8h\pi\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1} d\nu$$

$x = \frac{h\nu}{k_B T}$

$$= \frac{8\pi}{c^3} \frac{(k_B T)^4}{h^3} \int_0^\infty \frac{x^3 dx}{e^x - 1} = \frac{8\pi^5 k_B^4}{15c^3 h^3} T^4$$

Appendix J

The radiant energy current density (effusion 泻流)

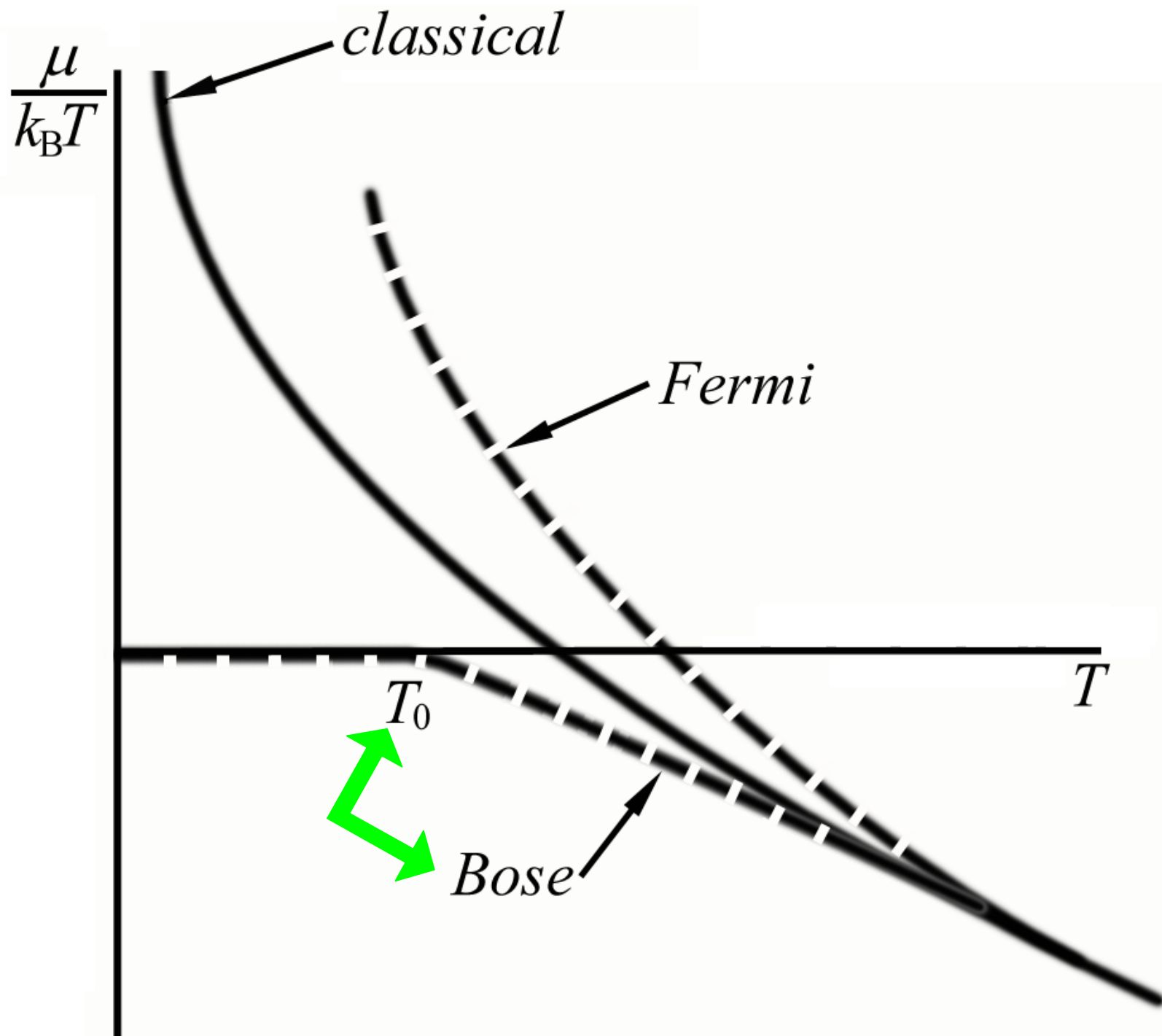
$$J_u = \frac{c}{4} u(T) = \sigma T^4. \quad \sigma = \frac{2\pi^5 k_B^4}{15c^2 h^3}.$$

Stefan-Boltzmann law

27.4 Bose-Einstein condensation

$$\left. \begin{array}{l} f_{FD}(E) \\ f_{MB}(E) \\ f_{BE}(E) \end{array} \right\} = \frac{1}{e^{\beta(E-\mu)} + \delta}, \quad \delta = \begin{cases} 1 & \text{Fermi - Dirac} \\ 0 & \text{Maxwell - Boltzmann} \\ -1 & \text{Bose - Einstein} \end{cases}$$

At low temperature, the Bose-Einstein distribution tends to emphasize the lower levels more than the Maxwell-Boltzmann distribution does, due to the negative sign in the denominator.



The total number of the particles(bosons) is

$$N = \sum_i n_i^B = \sum_i \frac{g_i}{e^{\beta(E_i - \mu)} - 1}$$

$$N = \int_0^\infty d\varepsilon \frac{g(\varepsilon)}{e^{\beta(\varepsilon - \mu)} - 1} = \frac{gV}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^\infty d\varepsilon \frac{\varepsilon^{1/2}}{e^{\beta(\varepsilon - \mu)} - 1}$$

At $T = T_0$, $\mu = 0$

$$\frac{N}{V} = \frac{g}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^\infty d\varepsilon \frac{\varepsilon^{1/2}}{e^{\varepsilon/k_B T_0} - 1}$$

$$\begin{aligned}
 \frac{N}{V} &= \frac{g}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^\infty d\varepsilon \frac{\varepsilon^{1/2}}{e^{\varepsilon/k_B T_0} - 1} \\
 &= \frac{g}{4\pi^2} \left(\frac{2mk_B T_0}{\hbar^2} \right)^{3/2} \int_0^\infty \frac{\sqrt{x} dx}{e^x - 1} \\
 &= \frac{g}{4\pi^2} \left(\frac{2mk_B T_0}{\hbar^2} \right)^{3/2} \zeta\left(\frac{3}{2}\right) \Gamma\left(\frac{3}{2}\right).
 \end{aligned}$$

$\frac{\varepsilon}{k_B T_0} \equiv x$


Thus

$$k_B T_0 = \frac{\hbar^2}{2m} \left[\frac{4\pi^2}{g \zeta\left(\frac{3}{2}\right) \Gamma\left(\frac{3}{2}\right)} \right]^{\frac{2}{3}} \left(\frac{N}{V} \right)^{\frac{2}{3}} = \frac{3.31}{g^{\frac{2}{3}}} \frac{\hbar^2}{m} \left(\frac{N}{V} \right)^{\frac{2}{3}}.$$

When $T < T_0$,

$$\begin{aligned}\frac{N_{\varepsilon>0}}{V} &= \frac{g}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^\infty d\varepsilon \frac{\varepsilon^{1/2}}{e^{\varepsilon/k_B T} - 1} \\ &= \frac{g}{4\pi^2} \left(\frac{2mk_B T}{\hbar^2} \right)^{3/2} \int_0^\infty \frac{\sqrt{x} dx}{e^x - 1} \\ &= \frac{g}{4\pi^2} \left(\frac{2mk_B T}{\hbar^2} \right)^{3/2} \zeta\left(\frac{3}{2}\right) \Gamma\left(\frac{3}{2}\right) \\ &= \frac{N}{V} \left(\frac{T}{T_0} \right)^{\frac{3}{2}}.\end{aligned}$$

$$\frac{\varepsilon}{k_B T} \equiv x$$

$$\frac{N_{\varepsilon=0}}{V} = \frac{N}{V} - \frac{N_{\varepsilon>0}}{V}$$

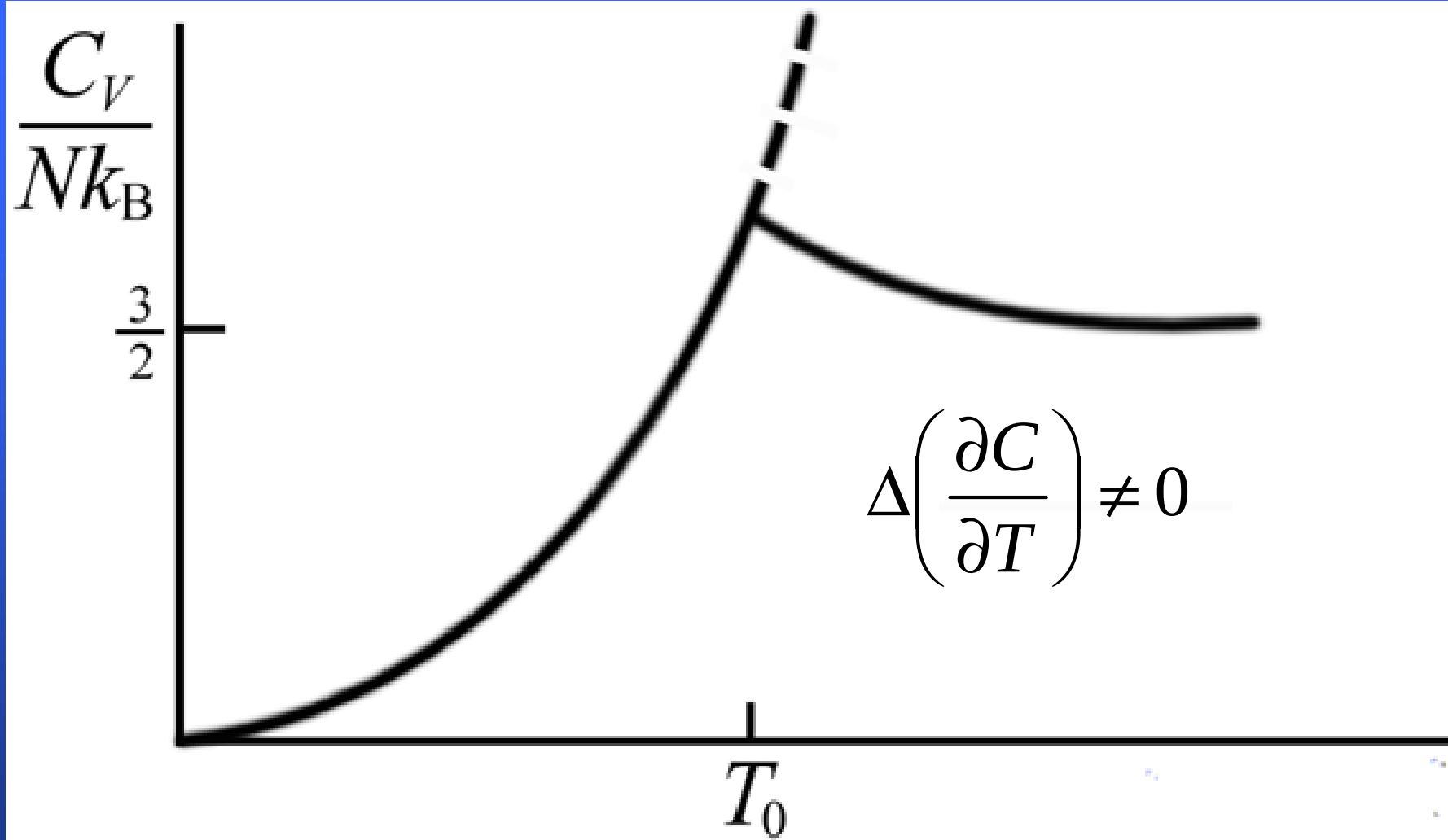
$$= \frac{N}{V} \left[1 - \left(\frac{T}{T_0} \right)^{3/2} \right]$$

A large number of particles begin to occupy the lowest energy state.-----Bose-Einstein condensation.

The particles on the lowest energy state-----
Bose-Einstein condensate.

Where do the particles condense to? $k=0$.

Condensation in the momentum space



^4He $T_0=3.14\text{K}$
 Phase transition of third
 order at 2.2K

$$C \sim T \frac{\partial S}{\partial T}; S \sim -\frac{\partial \mu}{\partial T}$$

In July 1995 a group in Boulder Colorado headed by C. Wieman and E. Cornell
cloud of ^{87}Rb atoms, 170nK.

In October, W. Ketterle of MIT and coworkers
a system of Na atoms, 2mK.

- rapid cooling

- high production rate of condensed atoms

- to take data about 30 times faster

- 300 times more BEC than Colorado group

2001 Nobel Laureates in Physics



Eric A. Cornell



Wolfgang Ketterle



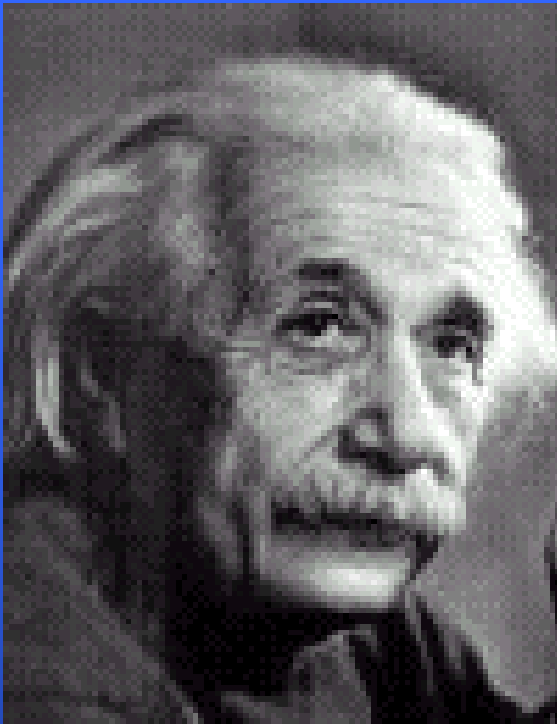
Carl E. Wieman

for the achievement of Bose-Einstein condensation in dilute gases of alkali atoms, and for early fundamental studies of the properties of the condensates

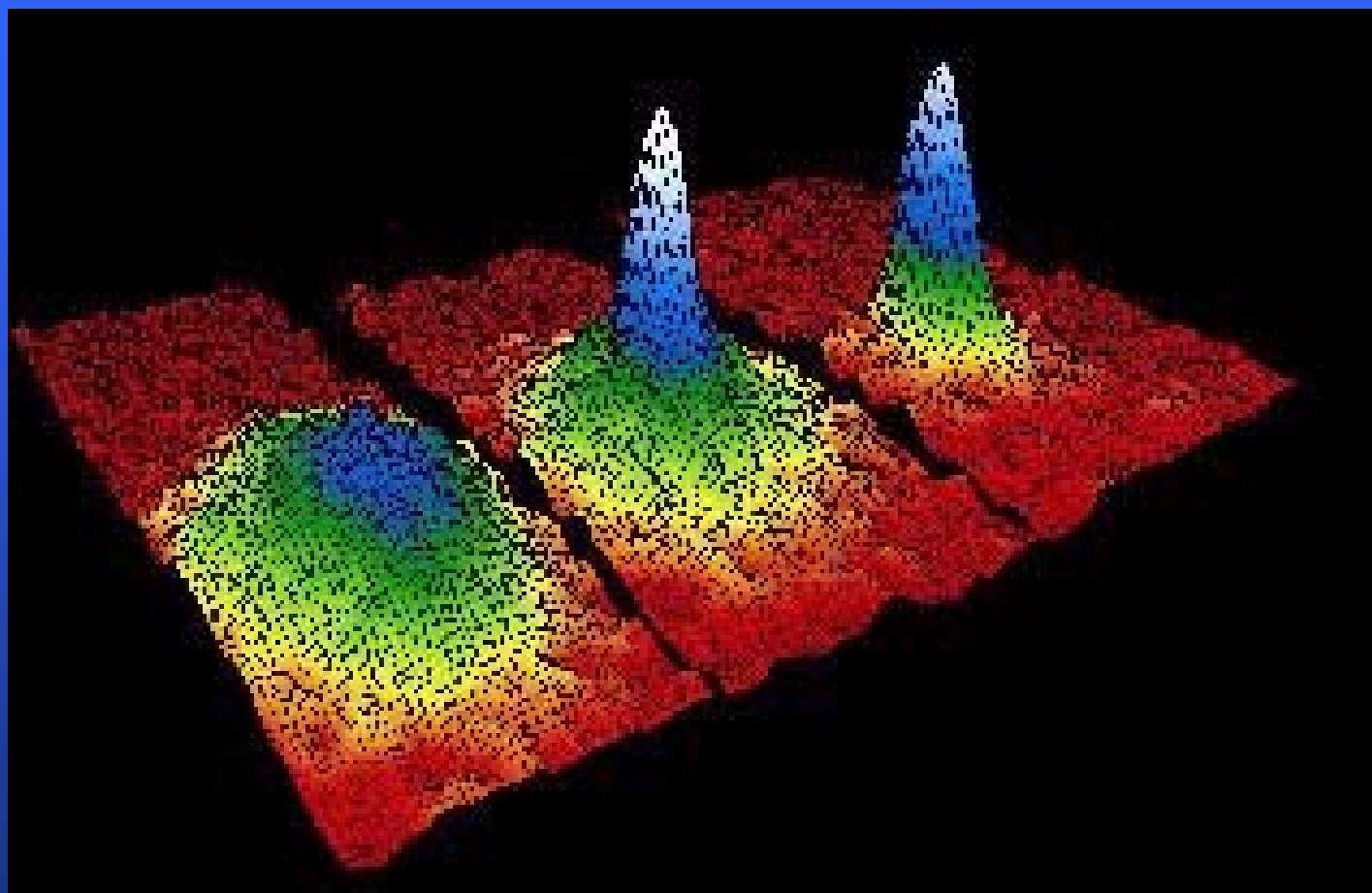
Eric A. Cornell 1961 - JILA and National Institute of Standards & Technology (NIST) , Boulder, CO, USA

Wolfgang Ketterle 1957- Massachusetts Institute of Technology (MIT) Cambridge, MA, USA

Carl E. Wieman 1951- JILA and University of Colorado ,Boulder, CO, USA



BEC Bose-Einstein Condensation

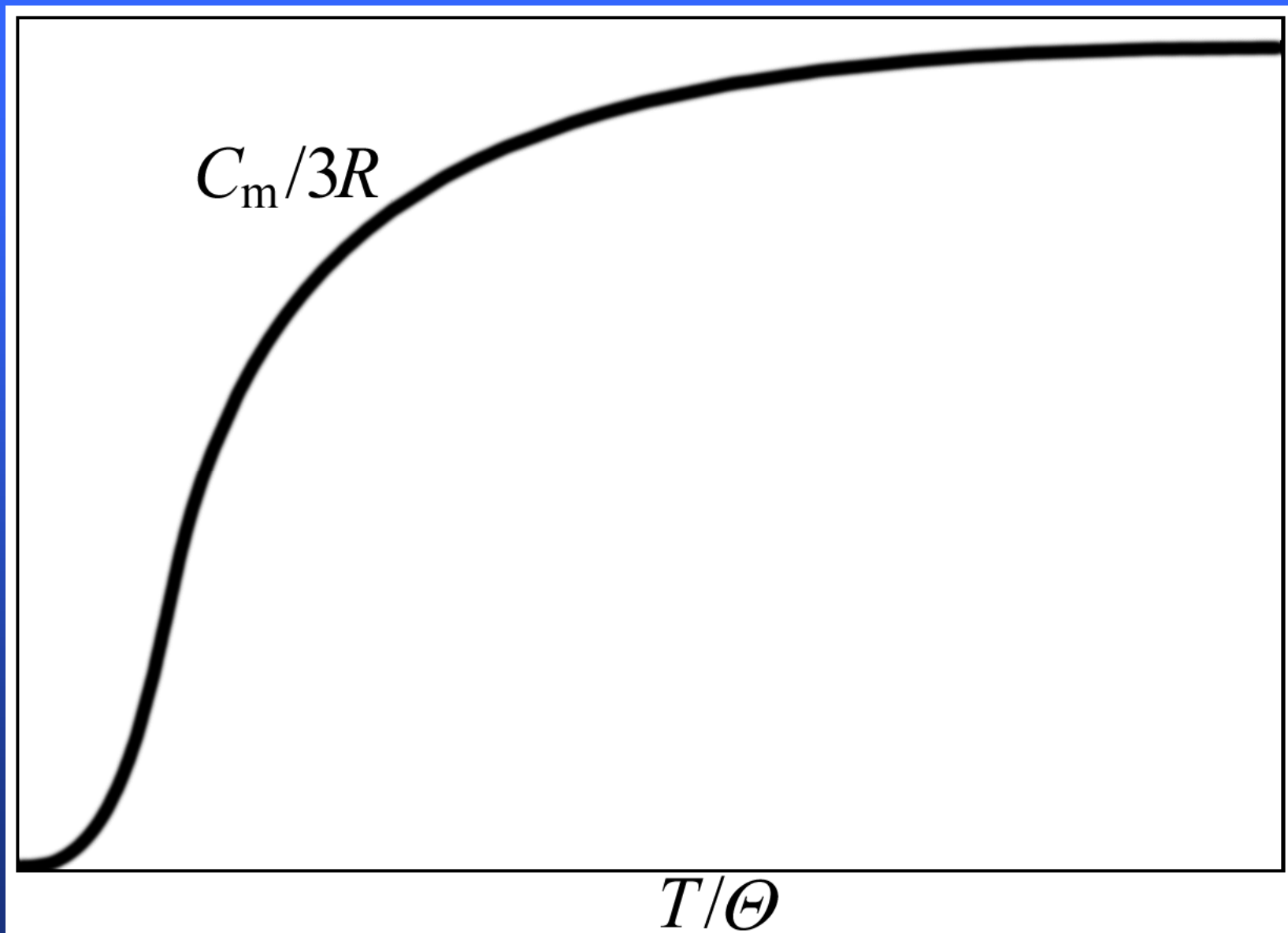


27.5 Specific heat of solids

Molar heat capacity: according to equipartition of energy

$$U \sim N_A \times 3 \times 2 \times \frac{1}{2} k_B T = 3RT$$

$$C_m = 3R \quad \text{Law of Dulong and Petit}$$



coupled oscillation and normal modes

Einstein assumed the energy of oscillations in solids was quantized. The quanta were called **phonons** and obeyed Bose-Einstein statistics

assumed that all phonons had same frequency. One mole atoms have $3N_A$ phonon modes, each with energy $\hbar\omega$.

$$E = 3N_A \hbar \omega \frac{1}{e^{\frac{\hbar\omega}{k_B T}} - 1}$$
$$C_m = 3R \left(\frac{\hbar\omega}{k_B T} \right)^2 \frac{e^{\frac{\hbar\omega}{k_B T}}}{\left[\exp\left(\frac{\hbar\omega}{k_B T} \right) - 1 \right]^2}$$

Debye assumed that phonons had a DOS

$$V \frac{\omega^2 d\omega}{2\pi^2} \left[\frac{1}{u_L^3} + \frac{2}{u_T^3} \right] = V \frac{\omega^2 d\omega}{2\pi^2} \frac{3}{\bar{u}^3}$$

$$V \frac{8\pi\nu^2 d\nu}{c^3} = V \frac{(2\pi)^3 \nu^2 d\nu}{2\pi^2} \frac{2}{c^3} \quad \text{photon}$$

Debye introduced a cut-off frequency—Debye frequency ω_D to prevent divergence of number of states.

$$V \frac{3}{2\pi^2 \bar{u}^3} \int_0^{\omega_D} \omega^2 d\omega = 3N$$

$$\omega_D^3 = 6\pi^2 \frac{N}{V} \bar{u}^3 \quad V \frac{\omega^2 d\omega}{2\pi^2 \bar{u}^3} \frac{3}{\omega_D^3} = 9N \frac{\omega^2 d\omega}{\omega_D^3}$$

$$U = U_0 + \frac{9N}{\omega_D^3} \int_0^{\omega_D} \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} \omega^2 d\omega$$

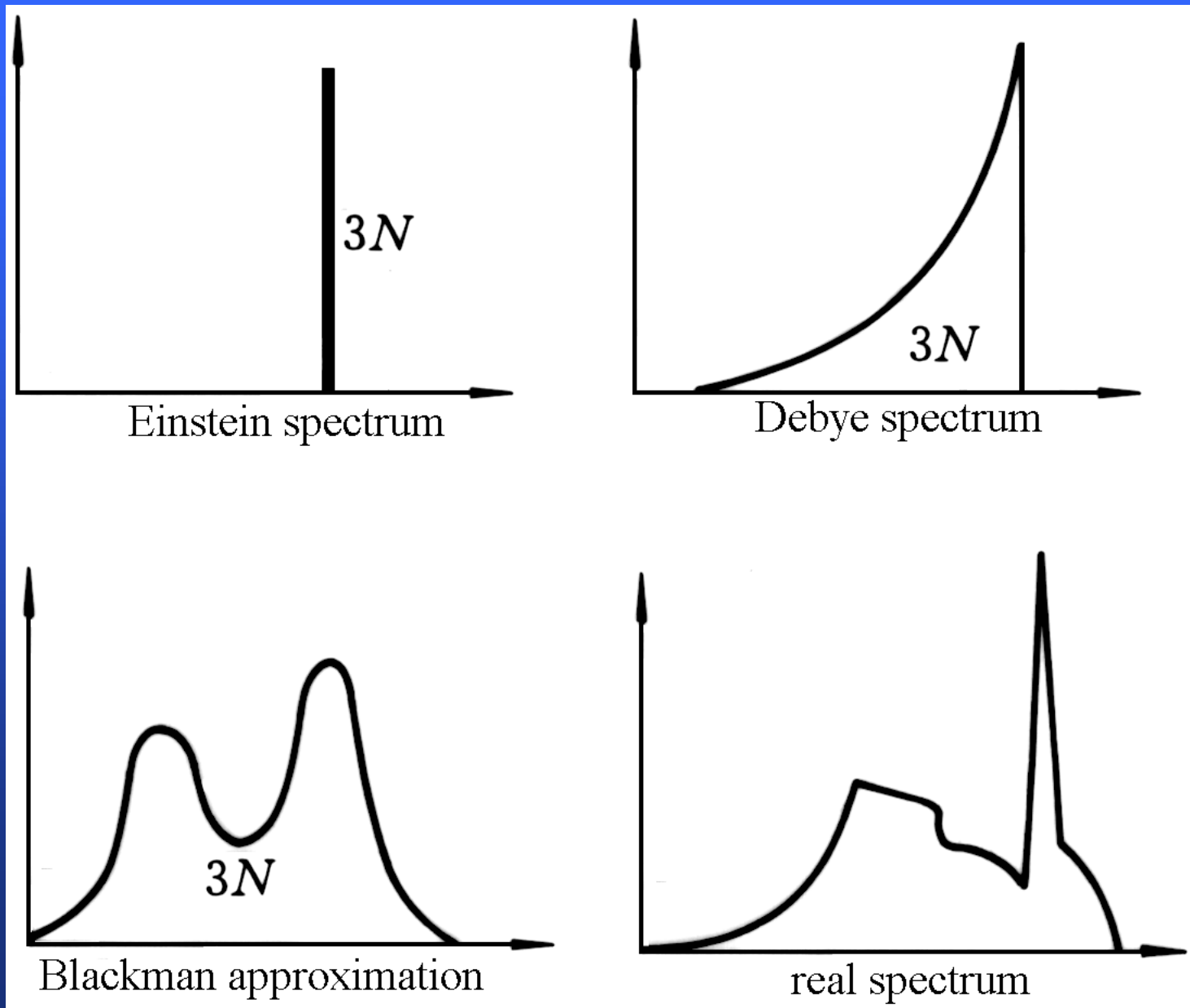
$$= U_0 + \frac{9Nk_B T}{(\Theta_D/T)^3} \int_0^{\Theta/T} \frac{x^3 dx}{e^x - 1}$$

$$k_B \Theta_D \equiv \hbar \omega_D$$

$$C_V = 3N k_B \left(\frac{T}{\Theta_D} \right)^3 \left[3 \int_0^{\Theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx \right]$$

$$C_V = \frac{12}{5} \pi^4 N k_B \left(\frac{T}{\Theta_D} \right)^3 \sim T^3 \quad (T \ll \Theta_D)$$

frequency distribution $\sim$ phonon spectrum



$$C_V = \alpha T^3 + \gamma T$$

contribution from electron

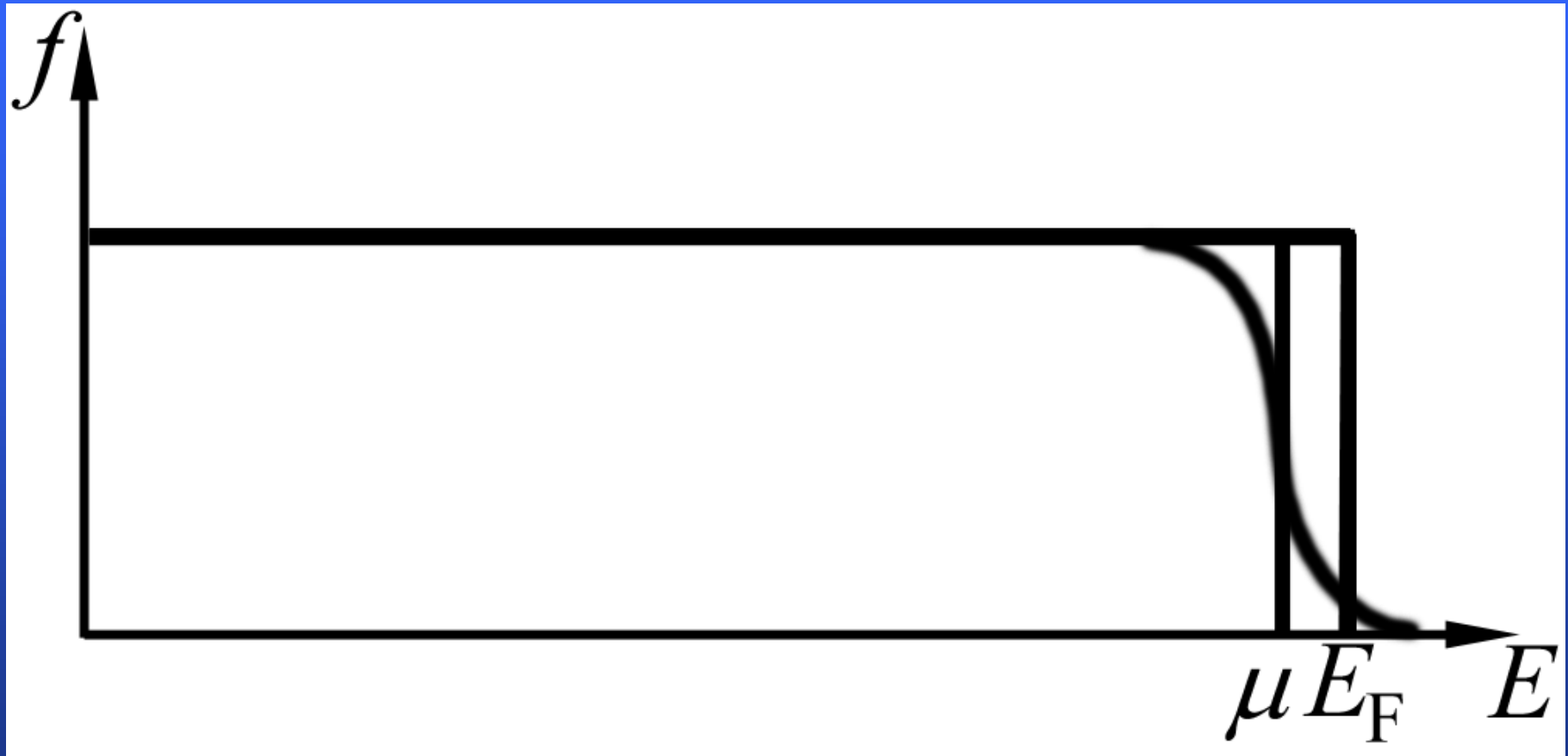
The number of states in the interval dE

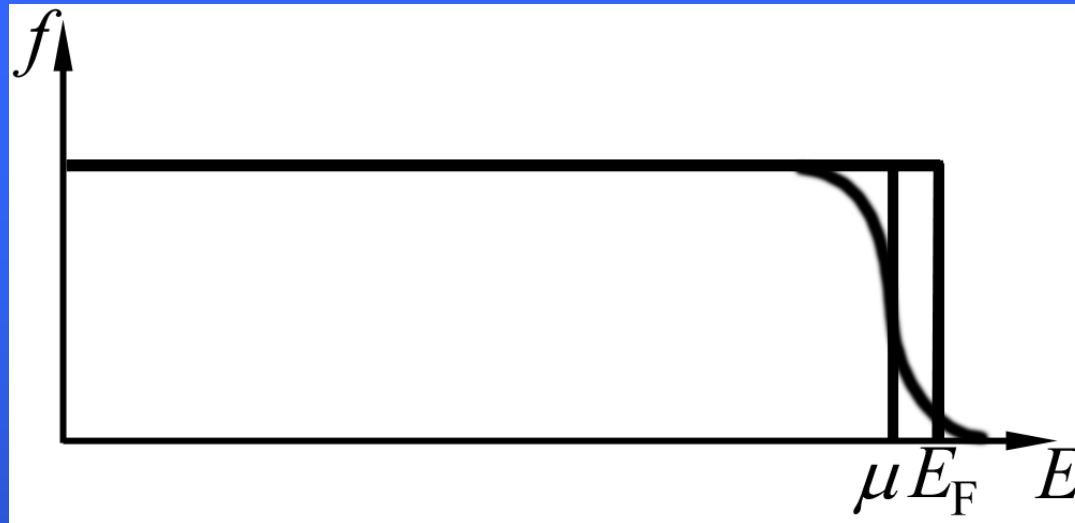
$$\rho(E) dE = \frac{3N}{2} \sqrt{\frac{E}{E_F}} \frac{dE}{E_F}$$

The number of electrons excited by thermal motion

$$N_e \sim \rho(E_F) \cdot k_B T = \frac{3N}{2E_F} k_B T = \frac{3}{2} N \frac{T}{T_F}$$

$$k_{\text{B}}T \sim 300\text{K} \sim 0.025\text{eV}$$





the energy involved in excitation

$$E_e \sim N_e \cdot k_B T = \frac{3}{2} N \frac{T}{T_F} \cdot k_B T$$

the specific heat is

$$\frac{C_V}{N} \sim 3k_B \frac{T}{T_F}$$

In 1900 Drude discovered that
(Chapter 16)

$$C_V = \frac{3}{2} \frac{n}{N_A} R = \frac{3}{2} n k_B$$

$$\kappa = \frac{1}{3} v_{\text{rms}} \bar{\lambda} n \left(\frac{3}{2} k_B \right)$$

$$\sigma = \frac{ne^2 \tau}{m} \approx \frac{ne^2}{m} \frac{\bar{\lambda}}{v_{\text{rms}}}$$

$$L = \frac{\kappa}{\sigma T} = \frac{3}{2} \left(\frac{k_B}{e} \right)^2 = 1.11 \times 10^{-8} \frac{\text{W} \cdot \Omega}{\text{K}^2}$$

$$j_e = -\sigma \nabla V$$

$$j_Q = -\kappa \nabla T$$

thermal conductivity

Correct result is

$$\frac{\kappa}{\sigma T} = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 = 2.45 \times 10^{-8} \frac{\text{W} \cdot \Omega}{\text{K}^2}$$

$$\left(\frac{C_V}{N}\right)_D \sim \frac{3}{2}k_B = \boxed{\frac{3}{2}k_B \frac{T}{T_F}} \frac{T_F}{T} \sim \frac{C_V}{N} \cdot 10^2$$

$$\left(\frac{1}{2}m\overline{v^2}\right)_D = \frac{3}{2}k_B T = \boxed{\frac{3}{2}k_B T_F} \frac{T}{T_F} \sim \frac{1}{2}m\overline{v^2} \cdot 10^{-2}$$

$\frac{1}{2}m\overline{v^2} \sim E_F = k_B T_F$

$$\frac{\kappa}{\sigma T} = \frac{2}{3} \frac{1}{2} m v_{\text{rms}}^2 \frac{\frac{3}{2}k_B}{e^2} \frac{1}{T}$$

Coulomb interaction vs free electron model

many-body effect $\sim$ quasi-independent

Assignment: 27.10